

Samuel E. Setsofia

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EDUCATION

Northwestern University | Evanston, IL

Anticipated December 2025

Master of Science in Artificial Intelligence

Deep Learning, Data Science, Human Computer Interaction, Language Models, Neural Networks, Generative AI

UNIVERSITY OF GHANA | Accra, Ghana

July 2020

Bachelor of Science in Biomedical Engineering,

WORK EXPERIENCE

NORTHWESTERN UNIVERSITY(Graduate Assistant) | Evanston, IL

September 2024 - Present

Assistant Educational Technology Specialist

- Served as a graduate assistant, facilitating generative AI workshops for 25 professors, boosting AI adoption in teaching. Designed user-centric course content in Canva and constructed Canva pages for effective teaching and learning

IRC | Rotterdam, Netherlands

October 2023 - August 2024

AI Chatbot Engineer Intern

- Built Knowledge Graphs for over 110 IRC's Water Sanitation and Health concepts/terminologies and fine-tuned LLAMA 2-7B model with RAG and semantic + vector search for chatbot development
- Devised chunking and tokenization strategies for over 100 WASH related research papers, storing embeddings in Chroma database

NYAHO MEDICAL CENTER | Accra, Ghana

October 2020 - January 2022

Biomedical Engineer Intern

- Trained over 120 physicians and health workers on medical ventilator use, reducing breakdown time by 50%. built a ReactJS-Django booking system and optimized data handling with web scraping and SQL, cutting equipment downtime by 30%

RESEARCH WORK

C3 Lab - Cognition, Creativity and Communication Lab | Evanston, IL

January 2025 - Present

Research Assistant

- Conducted research on the persuasive impact of LLMs on specific topics for 500+ users across U.S. ethnocentric geographies, analyzing cultural contexts, gender, political affiliations, and other factors

Reinforcement Learning agent for 'Oware-Namnam' board game |

December 2024 - January 2025

Evanston, IL

Personal Hobby Project

- Developed Double Deep Q-Network and Deep Q-Network agents for the 'Oware' board game, achieving a 90% performance improvement over DQN and random players. Enhanced gameplay strategy for Sub-Saharan African game

Northwestern University | Evanston, IL

October 2024 - December 2024

Machine Learning Project

- Constructed a reproducible data pipeline for modelling random forest and MLP for drug-target interaction prediction, optimizing drug discovery with molecular fingerprints extraction from 20,000 chemical compounds and protein targets

TECHNOLOGY SKILLS

- Python, R, JavaScript - ReactJS, SQL, DART - Flutter, Django REST Framework, HTML/CSS, A/B testing, Agile methodologies, MongoDB, MySQL, PostgreSQL, SQLite, Neo4j, Databricks, RESTful API, Postman, CI/CD, Git, UI/UX Design, Unit Testing, Docker, Conda, Kubernetes, Cloud computing platforms - Azure, AWS,
- Random Forests, Naive Bayes, KNN, SVM, Linear & Logistic Regression TensorFlow, Keras, Hugging Face, Scikit-Learn, PyTorch, Transformers, Autoencoders, CNNs, Deep Q-Networks, Time series analysis, Natural Language Processing (NLP),Pandas, NumPy, MS Excel, Matplotlib, Plotly, Power-BI, Scala, Tableau